
Fishora: Risk-Targeted Evaluation and Iterative Improvement of a Fish Identification and Evidence-Grounded Knowledge System

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Abstract

Fishora helps Indonesian fishermen identify their catch at landing and turns a confirmed species into an evidence-grounded knowledge card for buyers. Both halves fail in ways that aggregate metrics hide: a vision model can score 100% on curated photos yet misread real field photos, and a retrieval-augmented generator can produce fluent, plausible text that its evidence never states. We therefore built an evaluation infrastructure around the specific points where each component is most likely to break, locked every protocol before looking at results, and validated our automated judges against labelled data before trusting them. For computer vision, a 21,674-case suite spanning clean, corrupted, background-manipulated, real-field and out-of-distribution inputs showed that the baseline lost up to a third of its accuracy in the field, was confidently wrong on 21.3% of field photos, and accepted every non-fish image. Two iterations, photographer-level cross-fitted field adaptation and a two-stage rejection gate on DINOv3 embeddings, raised field accuracy of the deployable ViT-B from 67.1% to 86.0%, cut confident errors to 1.6%, rejected 100% of non-fish inputs, and ran 3.5× faster than the ViT-L MVP. For retrieval-augmented generation, a gold set of 221 locked claim labels showed that the baseline critic let 48% of unsupported claims through and that only 53% of shipped claims were supported. A claim-level verifier, a writer–critic workflow, a validated cross-family judge ($\kappa = 0.84$) and a new fact-coverage metric that cannot be inflated by sentence splitting raised supported claims to 100% of 79 judged claims and fact coverage from 44.2% to 87.1%, at \$0.69 per 1,000 cards. Every change is explained by the failure it targeted, including the changes we rejected.

Reading guide: where each required element and judging criterion is addressed.

Required element / criterion	Where
Test-suite coverage, case types, metrics and rationale	Section 4 (Tables 1–3), Section 4.4 automated tests
Why each test targets a risky point	Section 4.1, Section 4.2 (column "Risk it targets")
Why each metric, and what it does not capture	Section 4.3 (Table 3)
Baseline failure conditions with evidence	Section 5.1 (Tables 4–5), Section 6.1–6.2 (Tables 8–9)
Before vs. after, with causes	Section 5.2–5.3, Section 6.3–6.4, summary in Section 7 (Table 12)
Changes that failed or lowered a metric, and why	Section 8.1
Unresolved findings, causes, consistency with results, and plan	Section 8.2 (Table 13)
Dashboard	Section 9 (Figures 12–13)

1 Introduction

1.1 Background

Fish landed by small-scale fishers in Indonesia are frequently sold without reliable species information. Local names are ambiguous (the label *tenggiri* alone covers five scombrid species in FishBase), buyers have little verifiable information about taste, processing and uses, and fishers rarely have time to look it up at the dock. Fishora addresses this with two AI components (Figure 1): a **computer-vision (CV) identifier** that classifies a photo of the catch into one of 11 landed species and refuses inputs it should not classify, and a **retrieval-augmented generation (RAG) pipeline** that, once the fisherman confirms the species, writes an Indonesian knowledge card whose every claim is traceable to a verified source.

1.2 Why evaluation is the core of this work

Both components can look excellent under a naive test and still fail the fisherman. Our baseline DINOv3 models reached 100% accuracy on held-out dataset photos, yet the same models dropped to 67–82% on real field photos. Our baseline knowledge cards were fluent and well-formatted, yet only about half of their claims were stated by the evidence. We therefore treated the evaluation infrastructure as a first-class product: its job is to reveal *where* and *why* the system fails, so that every change can be justified by a specific failure and every improvement can be explained by its cause.

1.3 Contributions

1. **Risk-targeted test suites.** A 21,674-case CV suite (clean, 7 corruptions $\times$ 3 severities, background manipulation, 258 real field photos, 194 out-of-distribution images) and a RAG suite built on locked claim labels, hard negatives, multi-species and unresolved labels, and real-LLM runs with cost and latency.
2. **Validated measurement.** SHA-256 locked protocols, photographer-level cross-fitting, paired significance tests, bootstrap confidence intervals, and a judge-validation step that rejected one candidate judge before any system was scored.
3. **Explained iterations.** Two CV iterations and three RAG iterations, each tied to a named failure condition, with a quantified effect and a causal explanation, including the ideas that did *not* work.
4. **A metric that resists gaming.** We replaced "claims per card" with *fact coverage* after discovering that a prompt change could double the claim count while covering fewer facts.

2 Related Work

Self-supervised vision backbones. DINOv2 and DINOv3 learn general-purpose visual features without labels, so that a frozen backbone with a linear probe transfers well to new domains [1, 2]. We compare them with

supervised CNNs (MobileNetV3, EfficientNet-B0, ConvNeXt-T) [3–5].

Robustness, calibration and open-set rejection. Accuracy under common corruptions [6] and reliance on background shortcuts [7, 8] are well-known failure modes of image classifiers. Modern networks are often over-confident, which temperature scaling corrects [9]. Nearest-neighbour distance in feature space is a strong detector of unknown classes [10].

Grounded generation and its evaluation. RAG conditions generation on retrieved evidence [11], but citations alone do not guarantee support [12]. Atomic fact decomposition [13] and small entailment checkers [14] make faithfulness measurable at claim level. LLM judges must be validated before use [15], models cannot reliably self-correct without external feedback [16], and simple, well-measured workflows often outperform complex agent graphs at lower cost [17, 18]. Evaluation results need error bars [19].

3 System Overview

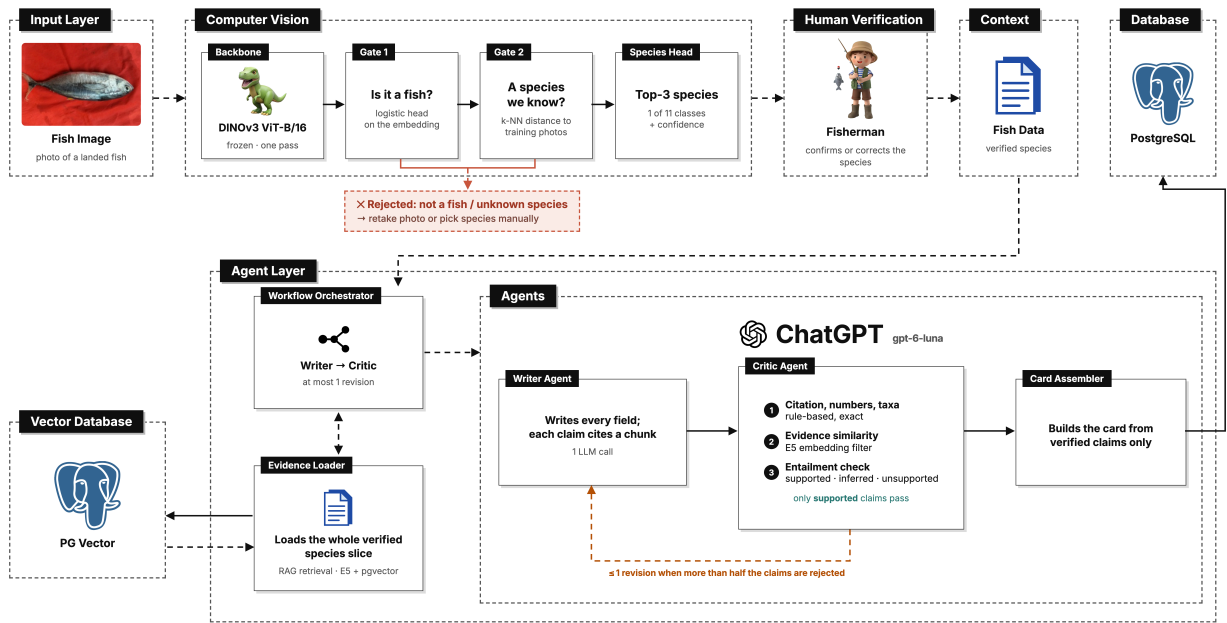


Figure 1: Fishora system architecture. The CV service and the knowledge pipeline share the verified species registry; the fisherman confirms the species before any card is generated.

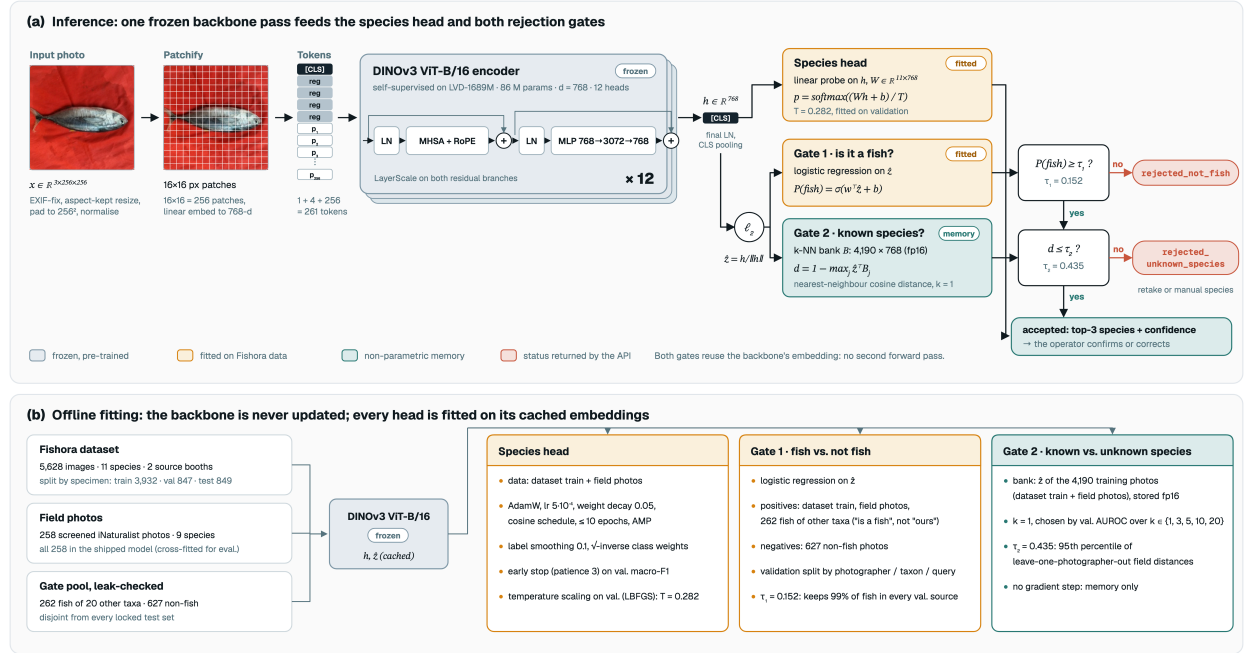


Figure 2: CV model architecture: frozen DINOv3 ViT-B/16, calibrated linear head, not-fish gate and unknown-species gate.

A fisherman photographs the catch in the Fishora web app. The CV service embeds the image with a frozen DINOv3 ViT-B/16, applies two rejection gates (not-fish, unknown-species), and returns a species with a calibrated confidence, or a rejection with a manual-selection fallback. After the fisherman confirms the species, a background job retrieves the species' verified evidence from PostgreSQL/pgvector using multilingual E5 embeddings [20], and a writer-critic workflow produces the knowledge card. Only claims that pass the verifier reach the card; fields without evidence stay empty instead of being guessed (Figure 1, Figure 2).

4 Evaluation Design

4.1 CV test cases and why they were chosen

Our guiding question was: *will the model work on a fisherman's photo, and will it refuse what it should not classify?* Table 1 lists the slices and the specific risk each one targets.

Table 1: CV test suite (21,674 cases per model).

Slice	n	Risk it targets
Clean held-out photos	849	Sanity check only; all backbones saturate here, so it is never the headline
Real field photos (iNaturalist, 11 species)	258	Domain shift from curated photos to boats and markets: wet fish, hands, ice, clutter
7 corruptions $\times$ 3 severities (blur, JPEG, low-res, dark, bright, rotation, occlusion)	17,829	Phone-camera conditions at the dock
Background removed / fish erased / background swapped	2,544	Whether the model looks at the fish or memorises the background
Unknown fish / non-fish / synthetic images	89 / 83 / 22	Open-set inputs: an 11-way classifier always answers unless it can refuse

Two controls make the field numbers trustworthy. First, all protocols (screening, field folds) are **SHA-256 locked** before results are computed. Second, when field photos are used to adapt the model, accuracy is **cross-fitted by photographer**: the 258 photos are split into two folds of 129 (87 and 101 photographers), and every photo is scored by a head that never saw that photographer.

4.2 RAG test cases and why they were chosen

Our guiding question was: *will a card state something its evidence does not say?* Hallucination in a knowledge card rarely looks like an obvious lie. It looks like a plausible inference (a buyer segment, a substitute species) or a fact that is true for a different species under the same label.

Table 2: RAG test suite.

Test	Risk it targets
221 gold claims from 44 real iteration-1 cards, labelled supported / inferred / unsupported against the species' verified evidence (AI-assisted annotation, reviewed by the team, SHA-256 locked)	Distinguishing a stated fact from a plausible inference
Hard negatives: a claim borrowed from another species in the same category	Copying a similar fish's fact into an empty field
Multi-species and unresolved labels: <i>tenggiri</i> (5 species), <i>tuna</i> (genus <i>Thunnus</i>), <i>gembolo</i> (no scientific identity)	Scope drift from one species to a whole label
Retrieval by species × category in Indonesian and English	Cross-species leakage and cross-lingual retrieval
Real-LLM runs on all 11 species with cost, latency and completion	Silent pipeline failures and operating cost
243 atomic evidence facts (locked)	Completeness of the card, independent of how claims are split

4.3 Metrics and what they do not capture

Table 3: Metrics, rationale and blind spots.

Metric	Why we use it	What it does not capture
Field accuracy (bootstrap CI)	Closest proxy to real use	iNaturalist photos approximate, but are not, our users' phones
Wrong & confident (≥ 0.9)	Confident errors are the ones a user trusts	Low-confidence errors
ECE	The app displays confidence	Unstable at small n
AUROC / FPR@95 for OOD	Whether unknown inputs are separable	Behaviour at the chosen threshold, so we also report pass rates
McNemar test	Whether a model difference is real	Only pairwise comparisons
CPU p50 latency, peak RAM	Deployability on modest hardware	Mobile or edge devices
Judge agreement with gold (κ , precision, recall, leak)	A judge must be validated before it scores systems	Errors shared by judge and gold
Supported-claim share	Faithfulness	Completeness: an empty card is 100% faithful
Fact coverage	Information content that cannot be inflated by sentence splitting	Knowledge absent from the corpus
Cost per 1,000 cards, p50/p95, completion	Operating viability	Indonesian fluency, buyer usefulness

4.4 Automated test suites and regression gates

Beyond the model-level suites, the repository contains **213 automated tests** that run offline with scripted LLM replies, so every behaviour below is re-checked on each change:

- **Retrieval quality gates** on the real corpus and the real E5 model: species purity of scoped retrieval = 1.0 (no cross-species leakage), every species × category cell reaches the top 6 (hit@6 = 1.0), the card query covers every available category, and warm query latency p95 < 150 ms.
- **Grounding tests** with hard negatives (a claim borrowed from another species in the same category) and cross-lingual pairs (an Indonesian claim grounded in English evidence).

- **Verifier and workflow tests:** unsupported numbers and taxa are rejected without an LLM call; a citation from another evidence category is rejected; a verifier outage fails closed; revision happens at most once; abbreviated binomials are accepted (regression test for the defect in Section 6.4).
- **Pipeline contracts:** the card cache follows the evidence, experts without evidence are not called, missing evidence is named on the card, and a card needs at most two sequential LLM rounds.
- **Corpus approval boundary:** only signed, human-approved chunks can be ingested; forged manifests and path-traversal attempts are rejected.

At baseline, 35 of 50 tests passed; after iteration 1, 71 of 71 passed; the current suite passes 213 of 213.

5 Computer Vision: Iterations and Results

5.1 Baseline: four specific failure conditions

We trained frozen-backbone linear probes on six backbones and ran the full suite (Table 4).

Table 4: Baseline linear probes (clean training data only).

Backbone	Clean acc.	Field acc.	Wrong & conf.	Background-swap flip	CPU p50
DINOv3 ViT-L (MVP)	100.0%	81.8%	15.1%	0.7%	896 ms
DINOv3 ViT-B	100.0%	67.1%	21.3%	0.1%	267 ms
DINOv3 ViT-S	99.6%	51.9%	38.8%	0.6%	97 ms
ConvNeXt-T	99.5%	53.1%	34.1%	10.7%	103 ms
EfficientNet-B0	94.9%	25.6%	20.9%	40.2%	53 ms
MobileNetV3-L	91.5%	29.8%	22.9%	38.1%	36 ms

The suite exposed four concrete failure conditions rather than a single low number:

1. **Domain gap.** Near-perfect clean accuracy collapsed on field photos (ViT-B: 100% $\rightarrow$ 67.1%). The linear head had only seen the photographic style of the training set.
2. **Over-confidence.** 21.3% of ViT-B field predictions were wrong *and* at least 90% confident, so a user would have trusted them.
3. **No refusal.** 100% of non-fish and synthetic images were accepted and labelled as a fish species.
4. **Background shortcuts in CNNs.** Swapping the background flipped 38–40% of MobileNet and EfficientNet predictions, versus at most 0.7% for DINOv3 models. The CNNs were partly classifying the scene.

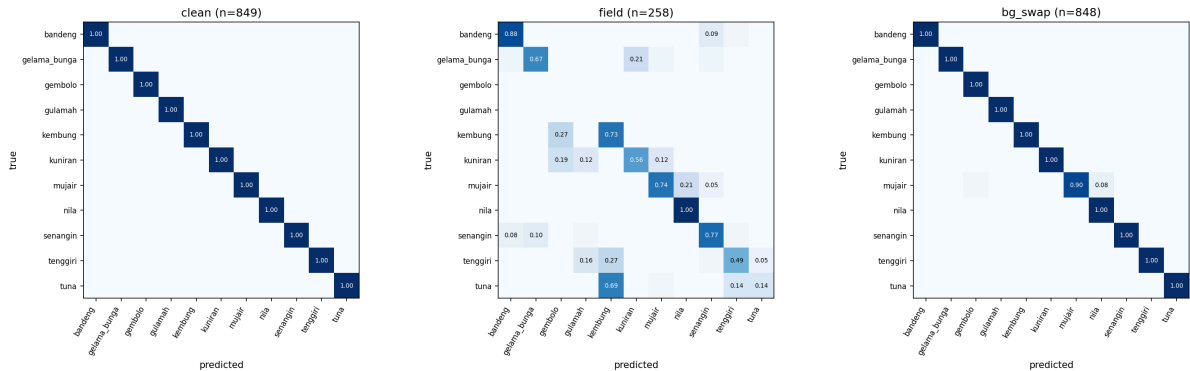


Figure 3: Baseline ViT-B confusion matrices. Clean held-out photos (left) and background-swapped photos (right) are almost perfectly diagonal, while real field photos (middle) scatter: tuna is read as kembung (0.69) and tenggiri as kembung (0.27). The failure is a domain gap, not a background shortcut.

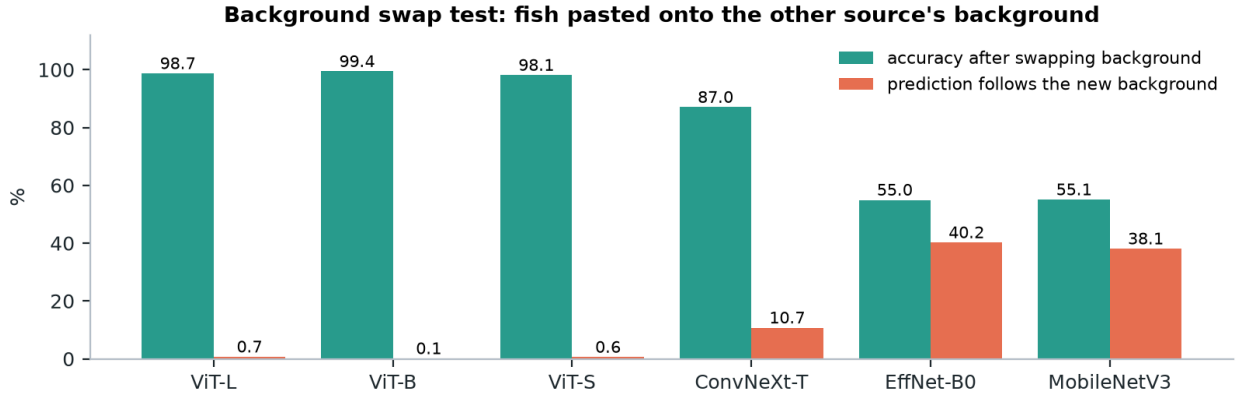


Figure 4: Background-swap test. DINOv3 predictions follow the fish ($\leq 0.7\%$ flips), while EfficientNet-B0 and MobileNetV3 follow the new background in 40.2% and 38.1% of cases.

Robustness at baseline. Image degradation hurt the smaller ViT-B much more than ViT-L (Figure 5): at blur radius 4 ViT-B kept 68.4% accuracy and at JPEG quality 8 about 62%, which motivated keeping corruption slices in every later evaluation.

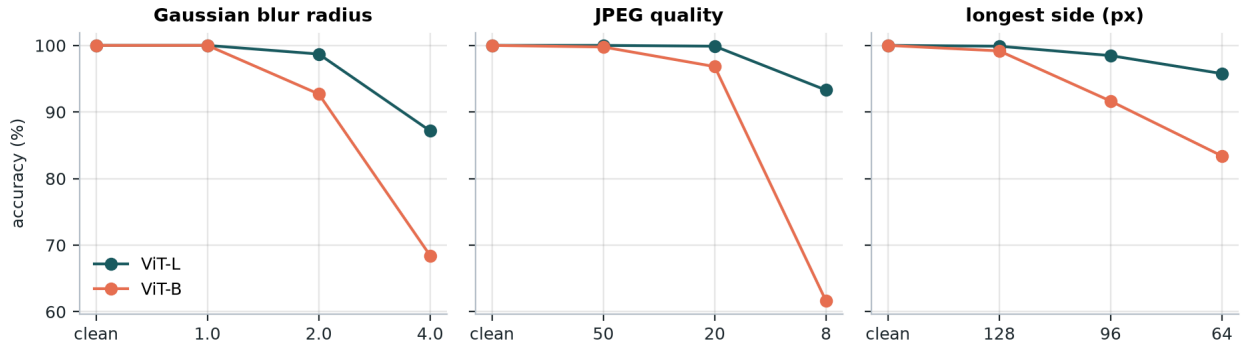


Figure 5: Baseline accuracy under increasing corruption severity (clean held-out photos). ViT-B degrades faster than ViT-L under strong blur, JPEG compression and low resolution.

Evidence at class level. The per-class field recall of the baseline ViT-B (Table 5) localises the domain gap. *Tuna* was recognised in only 13.8% of field photos, and 20 of its 29 photos were labelled *kembung*: both are scombrids with a fusiform body and finlets, and without field examples the head relied on silhouette. *Tenggiri* (48.6%) was confused with *kembung* for the same reason. By contrast, *nila*, whose vertical caudal stripes are distinctive, was already at 100%.

Table 5: Field recall per class, baseline ViT-B vs. after field adaptation (9 of 11 species have field photos).

Class	n	Baseline recall	Main baseline confusion	After adaptation	Main remaining confusion
tuna	29	13.8%	kembung (20)	75.9%	tenggiri (4)
tenggiri	37	48.6%	kembung (10)	81.1%	kembung (4)
kuniran	16	56.2%	gembolo (3)	87.5%	senangin (1)
gelama bunga	24	66.7%	kuniran (5)	87.5%	senangin (1)
kembung	22	72.7%	gembolo (6)	90.9%	gembolo (2)
mujair	19	73.7%	nila (4)	68.4%	nila (5)
senangin	48	77.1%	gelama bunga (5)	89.6%	gelama bunga (2)
bandeng	34	88.2%	senangin (3)	91.2%	tenggiri (2)
nila	29	100.0%	—	96.6%	senangin (1)

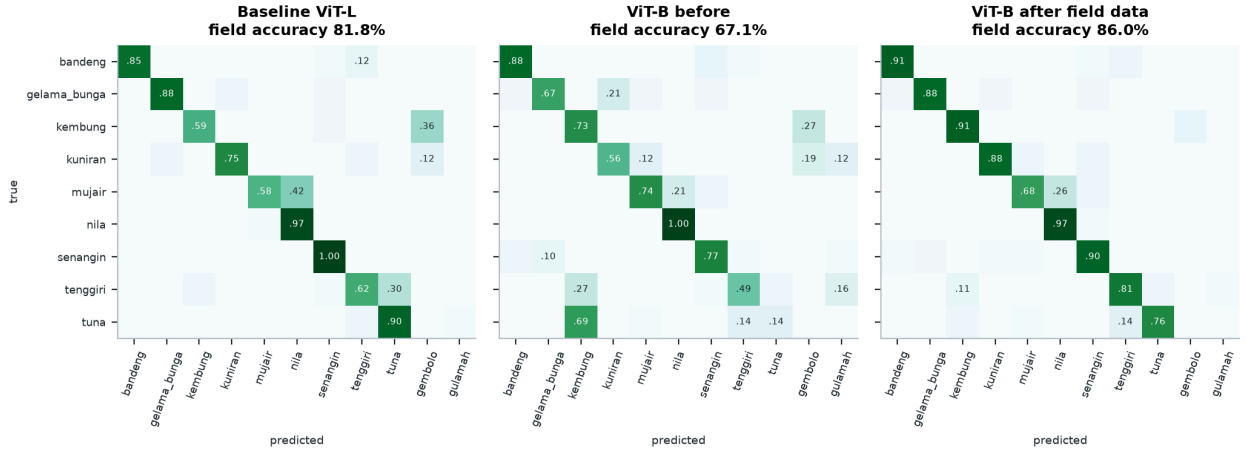


Figure 6: Field-photo confusion matrices (258 photos). Left: ViT-L MVP. Middle: ViT-B before field adaptation. Right: ViT-B after field adaptation (cross-fitted). The tuna $\rightarrow$ kembung and tenggiri $\rightarrow$ kembung confusions largely disappear; the remaining off-diagonal mass is mujair $\rightarrow$ nila.

5.2 Iteration 1: field adaptation with photographer-level cross-fitting

Change. We added field photos to the training of the linear head, scored with two-fold photographer cross-fitting (Section 4.1).

Result and cause. Every backbone improved, and the DINOv3 models improved most (Table 6): ViT-B rose from 67.1% to 86.0%, and confident errors fell from 21.3% to 1.6%. Because the backbone is frozen, the gain comes entirely from the head learning the field distribution of already-discriminative features. The small CNNs stayed below 36%, which confirms that their features, not their heads, were the bottleneck.

Table 6: Field accuracy before and after field adaptation (cross-fitted).

Backbone	Baseline	+ Field data	Δ	Wrong & conf. (after)	Unknown-fish AUROC (after)
DINOv3 ViT-L	81.8%	93.0%	+11.2	1.6%	0.940
DINOv3 ViT-B	67.1%	86.0%	+18.9	1.6%	0.952
DINOv3 ViT-S	51.9%	67.4%	+15.5	7.4%	0.946
ConvNeXt-T	53.1%	65.5%	+12.4	9.7%	0.884
MobileNetV3-L	29.8%	35.3%	+5.5	14.7%	0.753
EfficientNet-B0	25.6%	33.3%	+7.7	12.8%	0.737

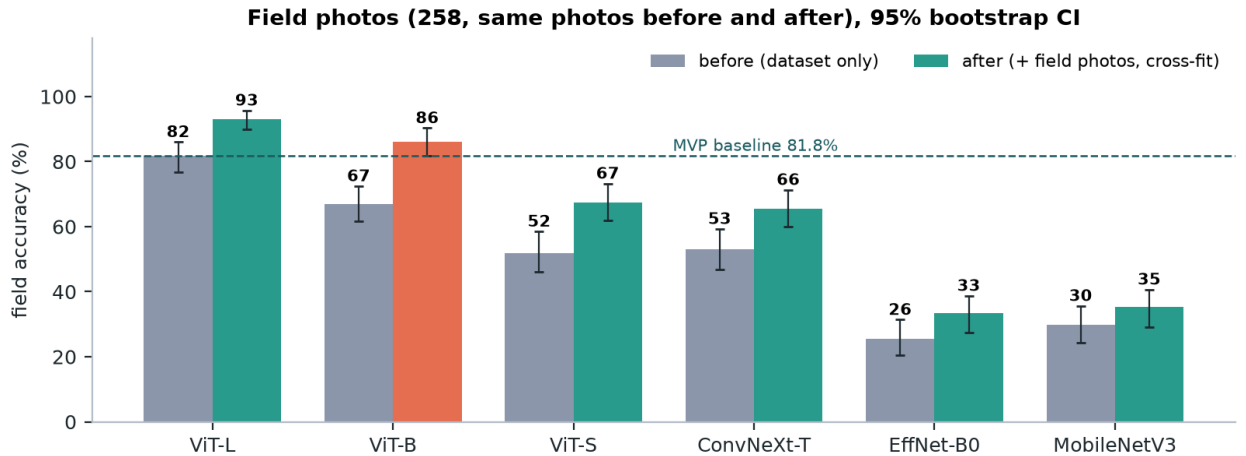


Figure 7: Field accuracy for all six backbones on the same 258 photos, before and after field adaptation (95% bootstrap CI). The dashed line marks the ViT-L MVP (81.8%).

Choosing the deployable model. ViT-L remains more accurate than ViT-B, and the gap is real: on paired field photos, ViT-L is right where ViT-B is wrong on 24 photos, versus 6 the other way (McNemar $p = 0.0014$). ViT-B, however, is $3.5\times$ faster (256 ms vs. 896 ms) and uses less than half the memory (1.35 GB vs. 2.96 GB peak RSS). For a service that must run on modest CPU hardware, ViT-B is the Pareto-optimal choice (Figure 8).

Field Accuracy vs. CPU Latency

Fishora CV · species accuracy on 258 real-world field photos (iNaturalist, 11 species) vs. CPU inference latency per image

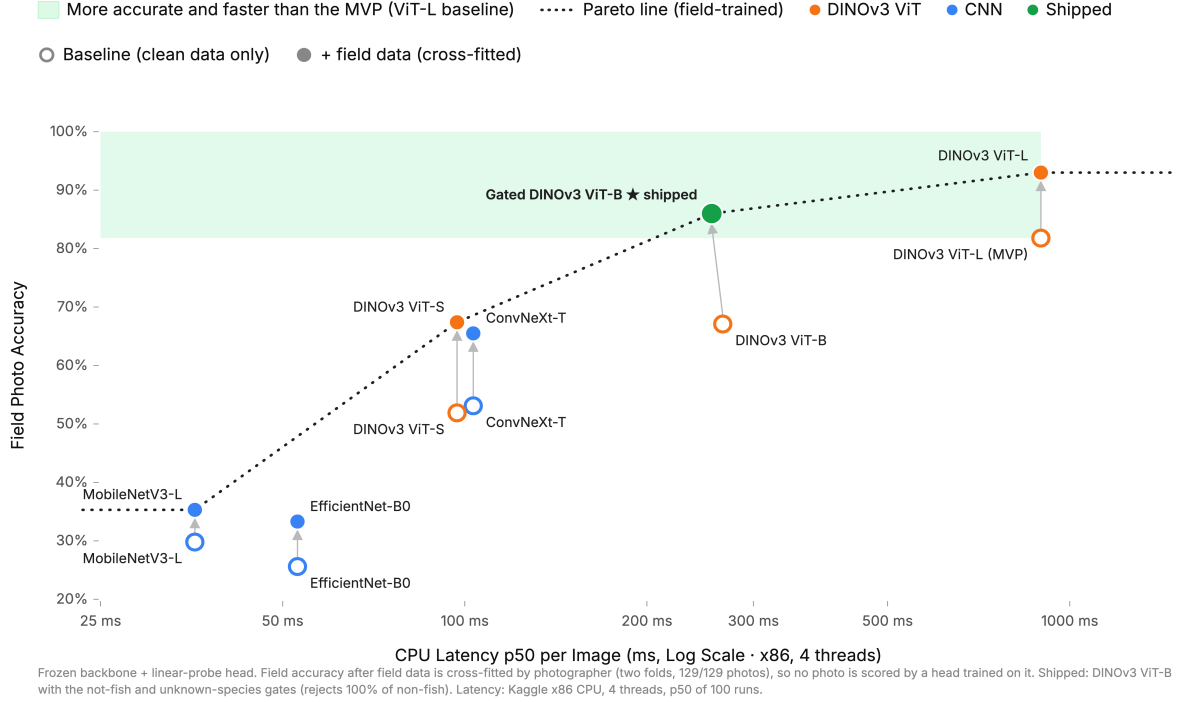


Figure 8: CV field accuracy versus CPU latency. Hollow points: baseline; filled: after field adaptation. The gated ViT-B is on the Pareto frontier and dominates the ViT-L MVP on both axes.

5.3 Iteration 2: calibration and a two-stage rejection gate

Change. We added (i) temperature scaling ($T = 0.282$) [9]; (ii) a *not-fish* gate, a logistic regression on the L2-normalised embedding ($\tau_1 = 0.152$); and (iii) an *unknown-species* gate, the cosine distance to the nearest of 4,190 training embeddings ($k = 1$, $\tau_2 = 0.435$) [10]. Rejected inputs return a clear status and a manual species selector instead of a wrong label.

Result and cause. Non-fish and synthetic images are now rejected 100% of the time (Table 7), because DINOv3 embeddings of non-fish images lie far from every fish class (AUROC 1.000). Robustness is preserved: the gated model keeps 100% accuracy on dark and rotated inputs, 97.1% under occlusion, and 89–94% under blur, JPEG and low resolution, and background swapping does not flip any prediction.

Table 7: Final gated DINOv3 ViT-B versus the MVP.

	MVP (ViT-L)	Baseline (ViT-B)	Final (gated ViT-B)
Field accuracy	81.8%	67.1%	86.0%
Wrong & confident on field	15.1%	21.3%	1.6%
Non-fish accepted	100%	100%	0%
Unknown fish rejected (AUROC)	0%	0%	32.6% (0.996)
CPU p50 latency	896 ms	267 ms	256 ms
Peak RAM	2.96 GB	1.34 GB	1.35 GB

The unknown-species gate is deliberately conservative. Unknown fish are separable (AUROC 0.996, FPR@95 = 2.2%), but at the production threshold 32.6% are rejected, because we prioritised keeping real field photos of the supported species: 96.5% of them are still accepted. Unknown fish such as other croakers and mackerels lie close to the known classes in embedding space, so the threshold trades rejections against false alarms (Section 8.2).

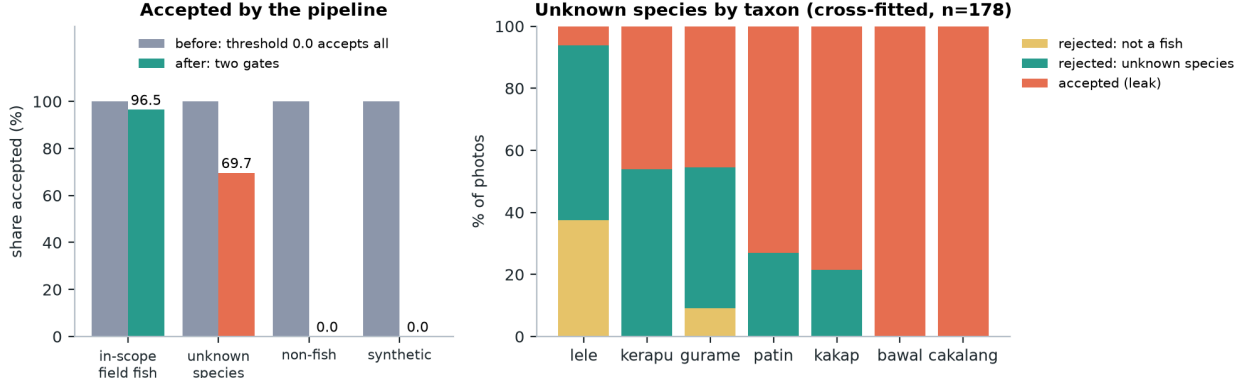


Figure 9: Left: share of inputs accepted before and after the gates. 96.5% of in-scope field photos are still accepted, while non-fish and synthetic images drop to 0%. Right: unknown-species leakage by taxon in the cross-fitted evaluation (178 photos, 69.7% accepted): species that resemble supported classes, such as cakalang (a scombrid) and bawal, leak most, while lele is almost always rejected.

6 Retrieval-Augmented Generation: Iterations and Results

6.1 Baseline and iteration 1: making the pipeline work

The baseline four-expert agent graph **completed 0% of real-LLM cards**: the provider rejected requests without a session header, and expert replies arrived as Responses-API content blocks or fenced JSON that the parser did not read. Iteration 1 fixed these defects and restructured the pipeline: the whole verified species slice (≤ 20 chunks) is used as evidence, each field is grounded only by its own evidence category, and LLM calls fell from 6.8 to 3.3 per card. Completion rose to **100%**, the test suite grew from 35/50 to 71/71 passing, and p50 latency fell from 6.0 s to 4.0 s. The critic became an E5 cosine filter ($\tau = 0.805$) that grounds Indonesian claims in English evidence.

6.2 Measuring the right thing: a locked gold set and a validated judge

Before changing the critic again, we built the measurement. We extracted all 221 unique claims from the 44 real iteration-1 cards and labelled each as *supported*, *inferred* or *unsupported* against the species' verified evidence, then locked the labels. Every candidate critic and judge was scored against them (Table 8).

Table 8: Graders scored against 221 locked gold claims. Leak = share of non-supported gold claims the grader accepts.

Grader	κ	Precision	Recall	Leak
Iteration-1 critic (E5 cosine)	0.07	0.70	0.65	48%
New verifier: deterministic + E5 + LLM entailment	0.26	0.88	0.64	15%
LLM entailment only	0.53	0.83	0.94	32%
Candidate judge: deepseek-v4.1-flash	0.50	0.77	0.91	45%
Selected judge: glm-5.3-flash	0.84	0.96	0.94	6%

Two baseline findings followed directly. First, the iteration-1 critic was near chance ($\kappa = 0.07$): cosine similarity measures *topic*, not *entailment*, so an inferred claim about the right species looks as similar as a stated one. Second, when the selected judge scored the real iteration-1 cards, only **53%** of their claims were supported;

most of the rest were *inferred*, typically buyer segments and substitute species, and the corpus contained **no substitute evidence at all**.

Evidence at claim level. The 82 non-supported gold claims are concentrated in three failure conditions (Table 9). *Buyer segments* account for 35 inferred claims: the evidence says milkfish is "marketed fresh, smoked, canned or frozen", and the card turned this into "buyers of fresh fish, buyers of smoked fish...". *Scope drift* appears on multi-species labels: evidence about *S. commerson*, one of five species called *tenggiri*, was restated as a fact about all *tenggiri*. *Substitutes* were filled with the other species that share the name *tenggiri*, which is a naming ambiguity, not a culinary substitution, and the corpus had no substitute evidence to ground them.

Table 9: Baseline failure conditions in the 221 gold claims (139 supported, 71 inferred, 11 unsupported).

Failure condition	Count	Example claim (card)	Why it is not supported
Invented buyer segments	35 inferred	"Pembeli ikan asap" (buyers of smoked fish), <i>bandeng</i>	Evidence lists product forms, never buyers
Scope drift on multi-species labels	20 inferred (processing, commercial)	"Dipasarkan segar" as a fact about all <i>tenggiri</i>	Evidence is about <i>S. commerson</i> only
Substitutes without substitute evidence	4 unsupported	" <i>Scomberomorus guttatus</i> " as a substitute for <i>tenggiri</i>	Shares the name; the corpus has no substitute chunk
Projection stated as fact	1 unsupported	Nila "contributes ... sales above USD 5 billion"	Evidence gives a projection, not a realised figure

6.3 Iteration 2: claim verifier, writer–critic workflow and model choice

Changes. (i) A claim verifier with three stages: deterministic checks (the cited chunk must be verified, belong to a category allowed for the field, and contain every number and taxon in the claim); an E5 filter; and one LLM entailment call per card that states a reason before the label. Only *supported* claims reach the card; a verifier outage fails the job instead of silently producing an empty card. (ii) A **writer–critic workflow**: one writer call sees all evidence and emits one claim per fact with citations, the verifier grades them, and a single bounded revision runs if more than half are rejected [17]. (iii) A 2×2 experiment, {4-expert graph, writer–critic} $\times$ {gpt-5.6-luna, gpt-6-luna}, with identical evidence, verifier and prompts, scored by the validated judge.

Table 10: Architecture $\times$ model experiment (11 species, real LLM, validated judge).

Configuration	Supported claims	Supported claims / card	Fact coverage	LLM calls	\$ / 1k cards	p50 / p95
Iteration-1 agent	53%	4.4	44.2%	3.3	0.90	4.0 / 67 s
4-expert · gpt-5.6-luna	98%	4.8	66.9%	4.2	1.81	10.6 / 72 s
4-expert · gpt-6-luna	96%	4.3	63.8%	4.2	0.79	7.5 / 12 s
Writer–critic · gpt-5.6-luna	100%	8.6	85.9%	1.9	1.73	14.1 / 18 s
Writer–critic · gpt-6-luna (shipped)	100% (79/79)	7.2	87.1%	1.9	0.69	10.3 / 14 s

Why the numbers moved.

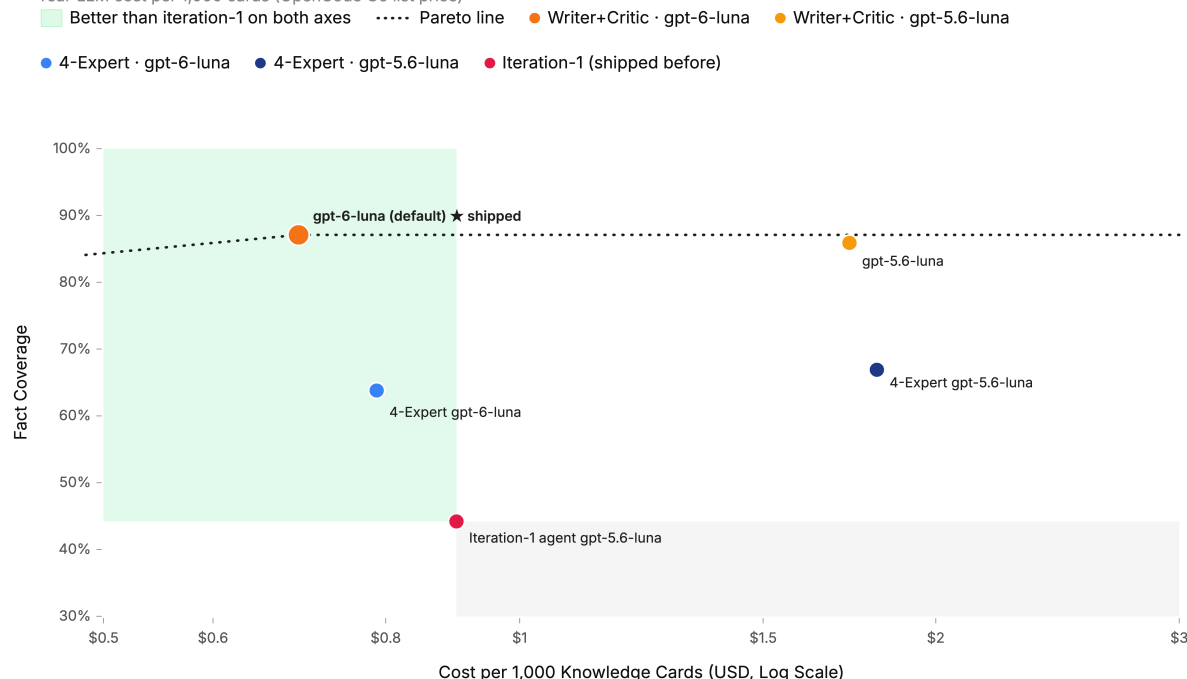
- **Verifier (53% $\rightarrow$ 96–100% supported).** Replacing topical similarity with claim-level entailment removes inferred claims. Leakage on gold fell from 48% to 15%.
- **Writer–critic ($\approx 1.7\times$ supported claims, 64–67% $\rightarrow$ 86–87% coverage, 4.2 $\rightarrow$ 1.9 calls).** One writer that sees all evidence at once does not lose facts at the boundaries between four narrowly scoped experts, and it needs fewer calls.

- **gpt-6-luna ($\approx 60\%$ cheaper).** Its list price is roughly half per token, and quality overlaps with gpt-5.6-luna.
- **p95 latency (67 s $\rightarrow$ 14 s).** Fewer sequential calls and retries.

The writer–critic workflow on gpt-6-luna is on the Pareto frontier of coverage versus cost and dominates the iteration-1 agent on both axes (Figure 10).

Card Architecture: Fact Coverage vs. Cost

Fishora RAG · share of locked atomic evidence facts stated by supported claims (judge: glm-5.3-flash) vs. real-LLM cost per 1,000 cards (OpenCode Go list price)



n = 11 cards per configuration (one per species, 163 atomic facts from the 49 corpus-v1 chunks); iteration-1: n = 22. Model default reasoning effort. Facts: AI-assisted annotation, reviewed by the team.

Figure 10: RAG fact coverage versus cost per 1,000 cards across architectures and models (11 species, 163 corpus-v1 facts). The shipped writer–critic workflow on gpt-6-luna is on the Pareto frontier.

6.4 Iteration 3: corpus v2, fact coverage and reasoning effort

Corpus v2. Fields without evidence stay empty by design, so card completeness is bounded by the corpus. We added 17 chunks with verbatim source quotes from FAO fact sheets and species catalogues and the Sydney Fish Market species guide, each scoped to the species it describes. The corpus grew from 49 to 66 chunks, covered species $\times$ category cells rose from 44 to 51 of 66, and **substitute evidence now exists for six species** where it previously existed for none.

Fact coverage as the headline metric. We decomposed the evidence into 243 atomic facts (locked) and measure the share that a card states with a supported claim [13]. Unlike claims per card, this metric does not reward splitting one sentence into several claims.

A verifier defect found by the new metric. Coverage for *tenggiri* was unexpectedly low. The cause was the deterministic taxon check: evidence wrote the abbreviated binomial "S. commerson" while claims wrote "Scomberomorus commerson", so correct claims were rejected. We now accept abbreviated binomials and family adjectives ("scombrid" for Scombridae), and added a regression test. Tenggiri's processing and substitute fields, previously empty, are now filled.

Reasoning effort. We swept the writer's reasoning effort on all 11 species with corpus v2 (Table 11; Figure 11).

Table 11: Reasoning-effort sweep, writer-critic on gpt-6-luna (11 species, 243 facts; 95% bootstrap CI over species).

Effort	Fact coverage	95% CI	\$ / 1k cards	p50
none	74.9%	64–84%	0.51	7.1 s
low	84.0%	80–91%	0.70	9.1 s
model default	77.0%	67–83%	0.96	13.6 s
medium	81.5%	72–89%	0.95	12.9 s
high	83.1%	77–92%	1.30	18.2 s
xhigh	81.9%	74–91%	1.94	24.4 s

Low effort is the best operating point: it has the highest coverage at 73% of the default's cost and a shorter latency. Coverage saturates above *low* because the bottleneck is the evidence, not reasoning: once the whole species slice is in context, additional reasoning mostly rephrases what is already stated.

Reasoning Effort: Fact Coverage vs. Cost

Fishora RAG · share of locked atomic evidence facts stated by supported claims (single LLM judge, Claude, same rubric for every card) vs. real-LLM cost per 1,000 cards (OpenCode Go list price)

■ Better than the model default on both axes Pareto line ● Writer+Critic · gpt-6-luna, by reasoning effort

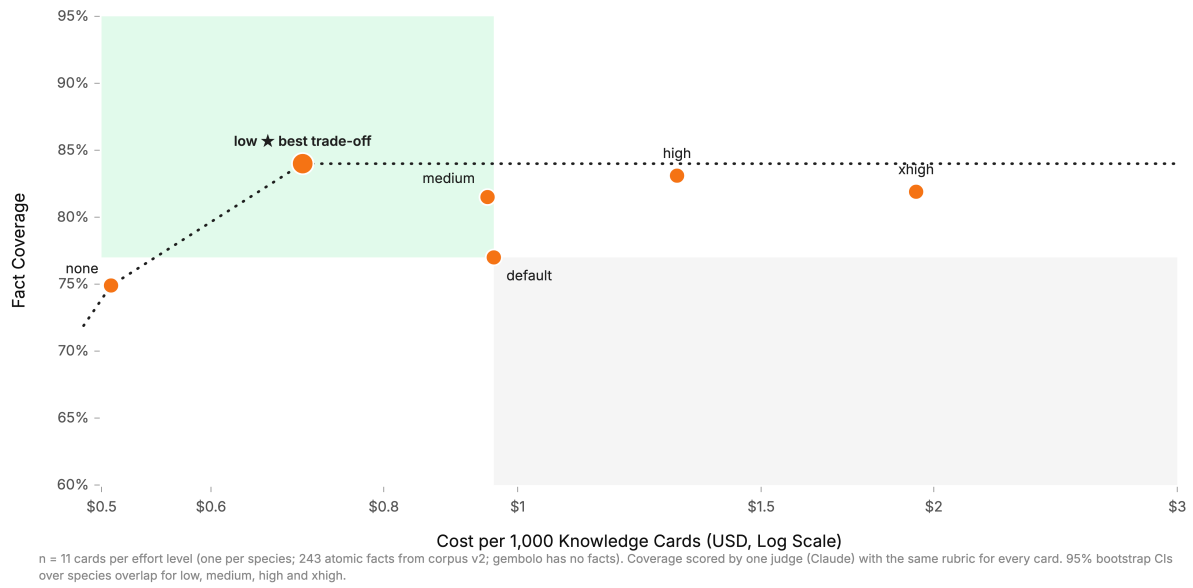


Figure 11: Reasoning-effort sweep on corpus v2 (11 species, 243 facts). Coverage saturates above low effort, which is the best operating point.

7 Before and After: Summary

Table 12: Before vs. after for both components, with the change responsible and its cause.

Component	Metric	Before	After	Responsible change	Why it moved
CV	Field accuracy (deployable model)	67.1%	86.0%	Field adaptation, cross-fitted	Head learns field distribution of frozen DINOv3 features
CV	Field accuracy vs. MVP	81.8% (ViT-L)	86.0% (ViT-B)	Same, on a smaller backbone	ViT-B features are sufficient once the head sees field data
CV	Wrong & confident on field	21.3%	1.6%	Field data + temperature scaling	Confidence calibrated on realistic inputs
CV	Non-fish accepted	100%	0%	Not-fish gate	Non-fish embeddings are far from all fish (AUROC 1.000)
CV	Tuna field recall	13.8%	75.9%	Field adaptation	Head no longer relies on scombrid silhouette alone
CV	CPU p50 latency	896 ms (MVP)	256 ms	ViT-B instead of ViT-L	3.5× fewer parameters
RAG	Real-LLM card completion	0%	100%	Iteration-1 pipeline fixes	Provider headers and reply parsing fixed
RAG	Supported claims	53%	100% (79/79)	Claim verifier	Entailment instead of topical similarity
RAG	Critic leak on gold	48%	15%	Claim verifier	Deterministic + entailment stages
RAG	Fact coverage	44.2%	87.1%	Writer–critic workflow	One writer sees all evidence
RAG	LLM calls per card	6.8	1.9	Iteration 1 + writer–critic	Fewer sequential rounds
RAG	p95 latency	67 s	14 s	Writer–critic	Fewer calls and retries
RAG	Cost per 1,000 cards	\$0.90	\$0.69	gpt-6-luna	Lower token price, same quality
RAG	Covered species × category cells	44 / 66	51 / 66	Corpus v2	17 new sourced chunks
RAG	Tests passing	35 / 50	213 / 213	All iterations	Every fix has a regression test

8 Analysis

8.1 Ideas that did not improve the system, and why

We report negative results because each one sharpened our understanding of the system.

- **Field adaptation slightly lowered *mujair* recall (73.7% → 68.4%) while raising every other scombrid and sciaenid class.** *Mujair* (*O. mossambicus*) and *nila* (*O. niloticus*) are congeneric tilapias; in field photos the distinguishing caudal stripes of *nila* are often not visible, and the adapted head, trained on more *nila* field photos, moved the shared boundary toward *nila*. Five of the six remaining errors are *mujair* → *nila*, consistent with this explanation.
- **"One atomic fact per claim" writer prompt (rejected).** It doubled claims per card (≈ 21) but covered fewer facts ($\approx 68\%$ vs. 87%) at 2.4× the cost. The prompt optimised the *count*: the writer spent its output on near-duplicate splits (1.9 claims per covered fact) instead of covering more evidence. This finding is why fact coverage replaced claims per card as our headline metric.
- **Higher reasoning effort.** Beyond *low*, coverage stays within the confidence interval while cost rises up to 2.8× (Table 11), because the evidence, not reasoning, is the limit.
- **The E5 filter lowers verifier recall** (0.94 → 0.64 relative to LLM-only). We keep it deliberately: it halves leakage (32% → 15%), and for a buyer-facing card omitting a true fact is safer than stating an unsupported one.

- **Retrieval recall@6 fell from 0.95 to 0.905 after corpus v2.** Top-6 retrieval selects one chunk per category first, so cells with several chunks cannot be fully recalled at $k = 6$. Card generation does not use this path (it reads the whole species slice), so we gate on $\text{hit}@6 = 1.0$ and document the reason.
- **Field data could not rescue small CNNs.** MobileNetV3 and EfficientNet stayed below 36% field accuracy; their background sensitivity (38–40% flips) shows the limitation lies in the features.

8.2 Unresolved findings: causes and plan

Each open item is explained by our own measurements (Table 13).

Table 13: Findings not yet resolved, with evidence, cause and plan.

Finding	Evidence	Cause (consistent with results)	Plan
Some unknown fish species pass the gate	67.4% of 89 unknown-fish images pass (69.7% in the cross-fitted 178-photo set); AUROC 0.996; leakage concentrated in cakalang and bawal (Figure 9)	Threshold chosen to keep supported field photos; unknown croakers and mackerels lie near known classes	Per-class thresholds; more unknown-fish reference images in the kNN bank
<i>Mujair</i> ↔ <i>nila</i> confusion	68.4% <i>mujair</i> recall; 5/6 errors → <i>nila</i>	Congeneric tilapias; distinguishing stripes often hidden in field photos	More <i>mujair</i> field photos; a dedicated tilapia sub-classifier on the same embedding
Field set is iNaturalist, not our users	258 photos, 9 of 11 species; <i>gembolo</i> and <i>gulamah</i> have none	Public field photos are unavailable for two local labels	Collect labelled photos from partner fishermen through the app's confirm/correct flow
<i>Gembolo</i> has no knowledge card content	FishBase returns no entry for the name	No resolvable scientific identity; any claim would be a guess	Expert taxonomic confirmation, then sourcing
15 of 66 species × category cells empty	e.g. taste for <i>gulamah</i> , <i>kembung</i> , <i>kuniran</i> , <i>senangin</i>	No open source with a verbatim quote found	Continue sourcing; the pipeline leaves these fields empty rather than inferring
Judge and verifier share a rubric	Judge $\kappa = 0.84$ on gold	Correlated errors are possible, so 100% supported is an upper bound	Second, independent judge family and periodic human audits
p50 card latency 4.0 s → 10.3 s	Table 10	Writer and verifier run sequentially	Acceptable: cards are generated in the background after confirmation; <i>low</i> effort brings p50 to 9.1 s

9 Evaluation Dashboard

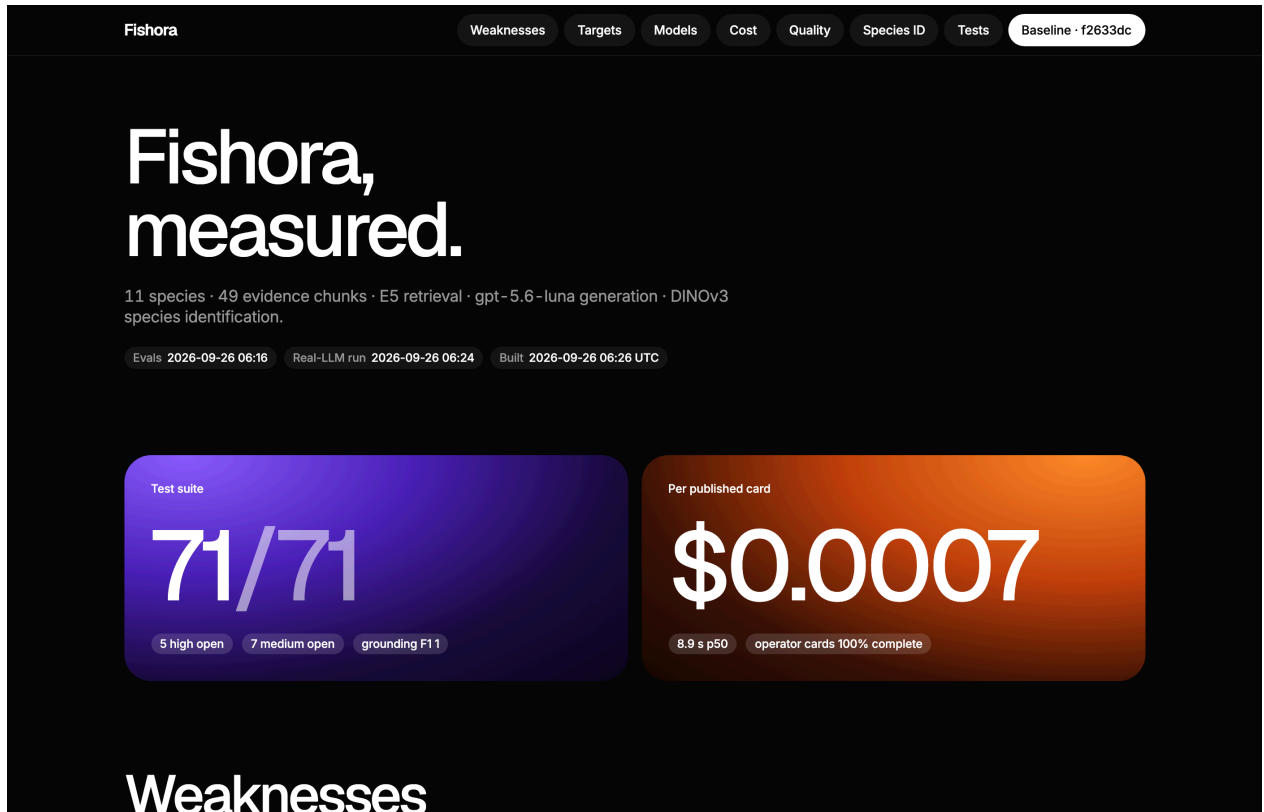


Figure 12: Evaluation dashboard (evals/results/iteration-1/dashboard.html): headline test-suite and cost figures.

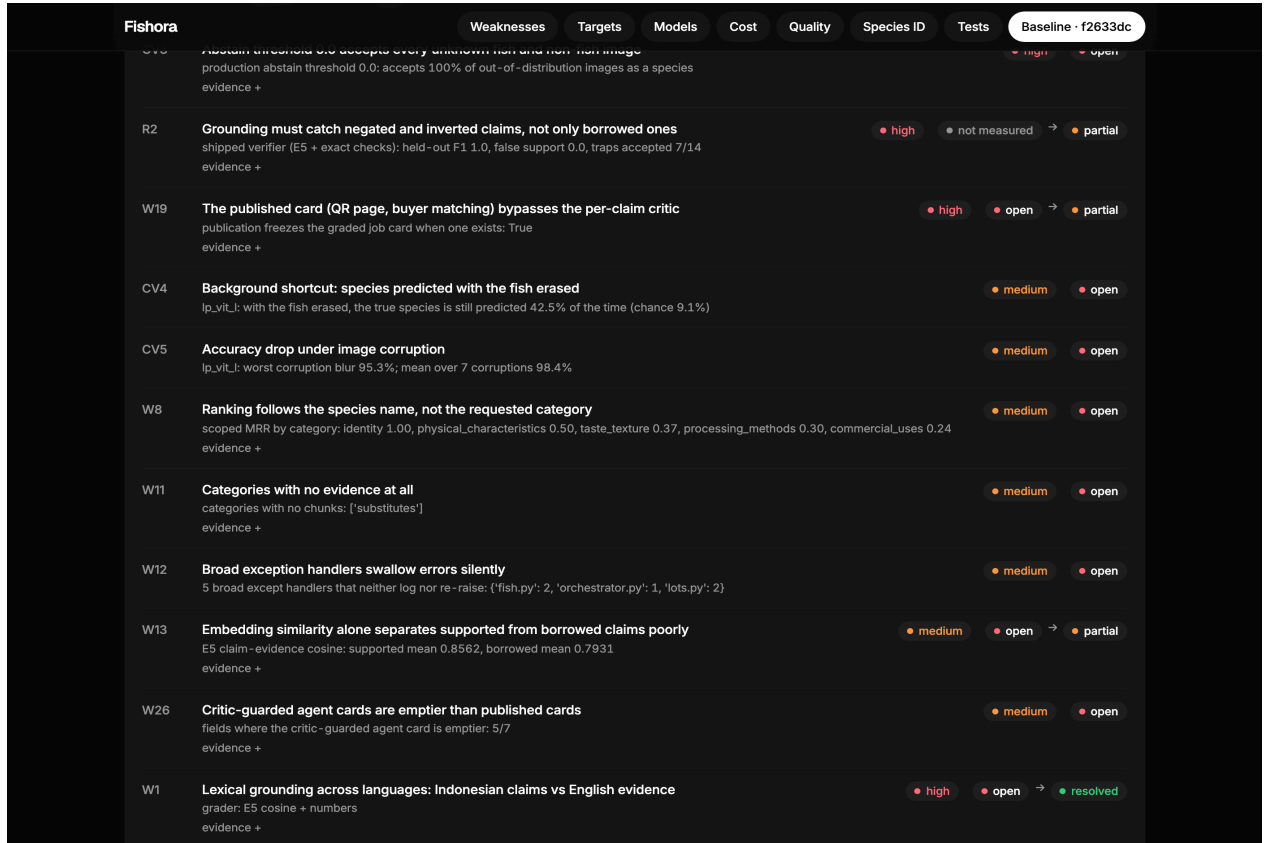


Figure 13: Dashboard weakness tracker: each finding carries a severity, its evidence and its status transition (open → partial → resolved).

The RAG evaluation is also published as a static dashboard generated from the run artifacts: `evals/results/iteration-1/dashboard.html` in the repository (open it in any browser). It lists every weakness with its severity, evidence and status transition (open $\rightarrow$ partial $\rightarrow$ resolved), the targets, model comparison, cost, quality, species-identification and test results (Figures 12–13). CV results are published per run in `evaluation/cv/results/<run>/` (`metrics.json`, `predictions.csv` and `figures`) and compared across runs in `evaluation/cv/results/compare/12_runs.csv`.

10 Reproducibility

All CV protocols (`evaluation/cv/protocol/*.lock`), the RAG gold claims (`evals/protocol/claims_gold.lock`) and the atomic facts (`evals/protocol/facts_gold.lock`) are SHA-256 locked, and the scripts refuse to run if a locked file changes. The test suites (213 tests) run offline with scripted LLM replies. Full CV evaluation runs on Kaggle (x86 CPU, 4 threads for latency); every number in this paper is produced by the scripts under `evaluation/cv/` and `evals/` and stored in `evaluation/cv/results/` and `evals/results/`.

11 Conclusion

By testing where Fishora is most likely to fail, validating our judges before trusting them, and changing our own headline metric when it proved gameable, we turned two components that looked strong on naive tests into components that are measurably reliable in the conditions that matter. The deployable CV model is 4.2 points more accurate than the MVP on real field photos, 3.5 $\times$ faster, confidently wrong on only 1.6% of field photos, and rejects every non-fish input. Knowledge cards went from 53% to 100% supported claims and from 44% to 87% fact coverage, at under one US dollar per thousand cards.

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